

LECTURE 9

System Design

When you "design" a system, you are essentially arranging these "Lego bricks" in a specific pattern:

Load Balancer (The Traffic Cop): When people arrive, a "Traffic Cop" stands at the front and points them to different lines. "You go to Stand 1, you go to Stand 2." This ensures no single stand gets overwhelmed.

Caching (The "Pre-made" Pile): Instead of squeezing a lemon every time someone asks, you keep 50 cups already poured and sitting on ice. It's much faster to hand someone a pre-made cup than to make it from scratch.

Database Sharding (The Filing Cabinets): If you have a list of a billion customers, it's too heavy for one book. You split it up: "Customers A-M are in this book; N-Z are in that book." Now, two people can look up names at the same time.

The "Pick Two" Rule (CAP Theorem)

In a perfect world, a system would be **Fast, Correct, and Never Break**. But in reality, you usually have to pick two:

1. **Consistency**: Everyone sees the same thing at the exact same time.
2. **Availability**: The site is always "on," even if it's a little glitchy.
3. **Partition Tolerance**: The system keeps working even if the internet cables are cut in half.

Example: *On Facebook, it's okay if you see "10 Likes" and your friend sees "12 Likes" for a few seconds. Facebook chooses Availability (the site stays on) over Consistency (perfectly identical numbers every millisecond).*

Systems in Product Development

Product is a Living System

- Technology
- User Behavior
- Market
- Regulations
- Data
- Ethics

Systems in Product Development

System Failures in Products

- **Over-optimisation** - (*Optimising one metric so much that the overall system suffers*)
- **Ignoring stakeholders** - (*Designing for primary users while neglecting secondary or invisible actors*)
- **Lack of feedback loops** - (No mechanism to learn, adapt, or self-correct)

Systems in Product Development

Over-optimisation

Over-optimization happens when a product team focuses intensely on one KPI (speed, engagement, profit, battery life, etc.) and ignores the holistic impact.

System Failures in Products

Over-optimisation - Example 1

Smartphone Battery Life vs Performance

What was optimised: Battery endurance

System Failure:

- Aggressive background app restrictions.
- Notifications delayed.
- Users missed important messages.
- Perceived product “lag” or “bugs.”

System Insight: Battery metric improved, but user trust and usability decreased.

Systems in Product Development

Ignoring stakeholders

Products rarely serve just one group. Ignoring stakeholders creates friction and resistance.

System Failures in Products

Ignoring Stakeholders - Example 1

Smart City Surveillance Cameras

Primary Focus: Safety & crime reduction

Ignored Stakeholders: Citizens' privacy advocates, civil liberty groups

System Failure:

- Legal challenges.
- Public distrust.
- Policy restrictions.
- Reduced adoption

System Insight: Security was optimised, but privacy ecosystem backlash destabilised the initiative.

System Failures in Products

Ignoring Stakeholders - Example 2

Electric Scooter Sharing

Primary Focus: Urban commuters

Ignored Stakeholders: Pedestrians, city maintenance teams, elderly citizens

System Failure:

- Scooters blocking sidewalks.
- Accidents.
- City bans or strict regulations.

System Insight: The physical environment and non-users are also stakeholders.

Systems in Product Development

Lack of Feedback Loops

When products don't capture or act on feedback, they drift away from real needs.

System Failures in Products

Lack of Feedback Loops - Example 1

Enterprise Software Without User Analytics

Issue: Product updates based on assumptions, not usage data

System Failure:

- Features nobody used.
- Important needs ignored.
- Costly redesigns later.

System Insight: Without usage loops, design decisions become guesswork.

System Failures in Products

Lack of Feedback Loops - Example 2

Fitness Wearables Without Behavior Feedback

Issue: Only data display, no coaching or insights

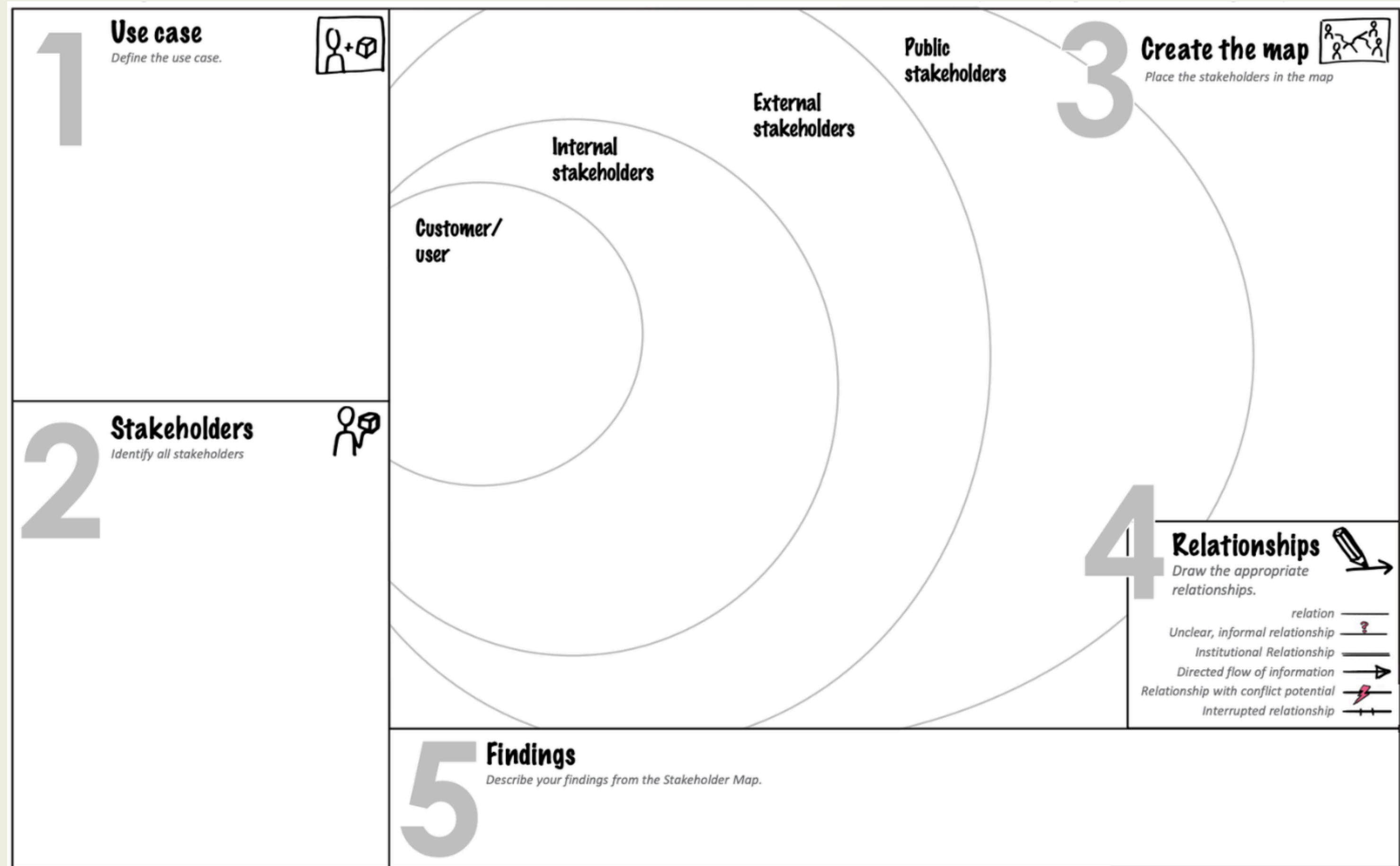
System Failure:

- Users stopped wearing devices.
- Data overload without action.
- High drop-off rates

System Insight: Information without interpretive feedback doesn't change behavior.

Systems in Product Development

Tools : Stakeholder Map



Systems in Product Development

Tools : Causal Loop Diagram

Causal Loop Diagram

A Causal Diagram (also called a Causal Loop Diagram – CLD) is a visual map that shows how different variables in a system influence each other.

Instead of asking

“What caused this?”

it asks

“What causes what — and how do they all connect?”

It helps us see chains and circles of cause–effect relationships, not just one–way arrows.

Core Elements of a Causal Diagram

1. Variables

Things that can change in the system.

Examples: Users, Revenue, Stress Level, Traffic, Battery Life.

2. Arrows (Causal Links)

Show influence between variables.

3. Polarity Signs

- (+) Positive Link → Move in the same direction
- If A increases, B increases. If A decreases, B decreases.
- (–) Negative Link → Move in the opposite direction
- If A increases, B decreases.

4. Loops

When arrows form a circle, we get feedback loops:

- Reinforcing (R) → Growth or decline snowball
- Balancing (B) → Stabilizing or controlling effect

Example of a Causal Diagram

Mobile App Notifications

1. Variables

- Notifications Sent
- User Engagement
- User Annoyance
- App Uninstalls

2. Diagram Logic

Notifications (+) → Engagement (+)

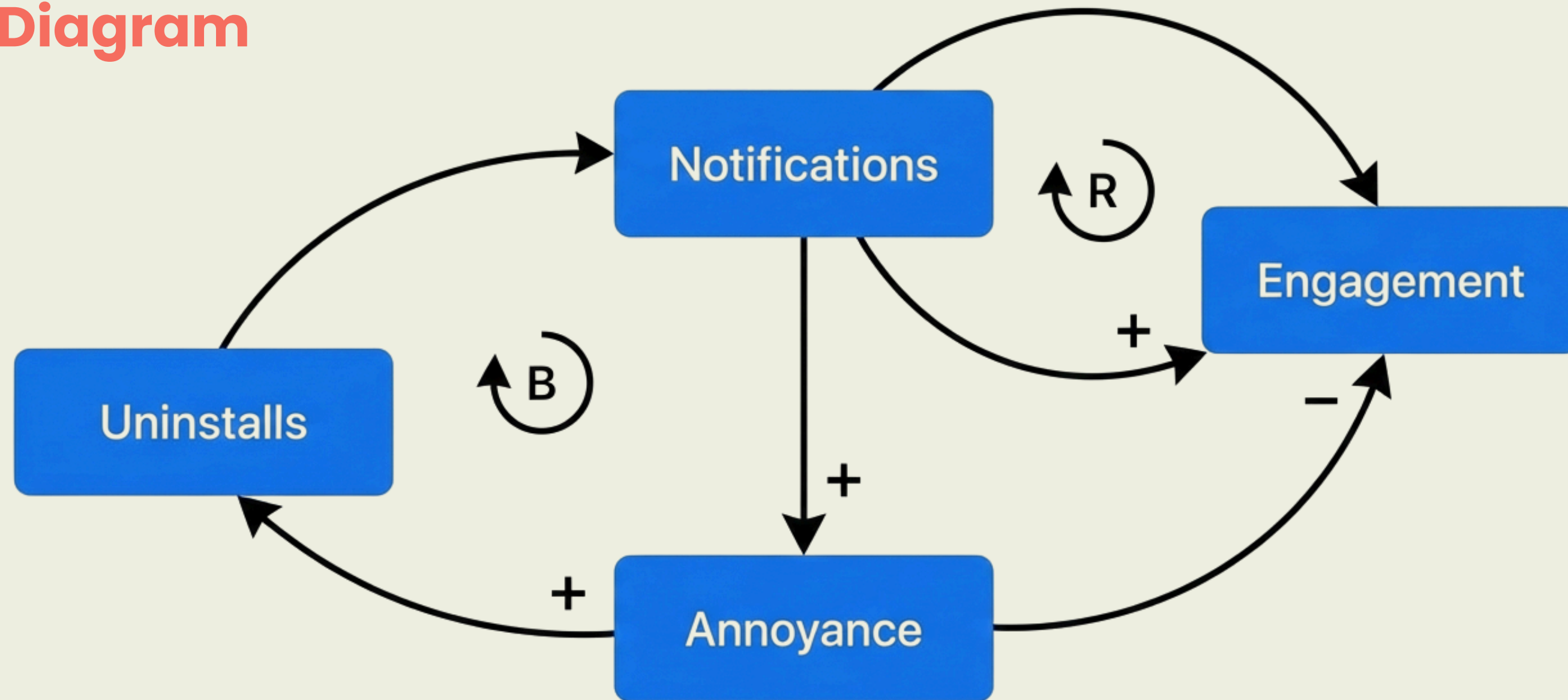
Notifications (+) → Annoyance (+)

Annoyance (+) → Uninstalls (+)

Uninstalls (−) → Engagement

Example of a Causal Diagram

Mobile App Notifications Diagram



Example of a Causal Diagram

Mobile App Notifications

3.Insight

- Sending notifications boosts engagement initially
- But also increases annoyance
- Annoyed users uninstall
- Engagement drops later

A causal diagram reveals unintended consequences that linear thinking misses.

Example of a Causal Diagram

Learning & Skill Improvement (Reinforcing Loop)

Variables

- Practice Time
- Skill Level
- Confidence
- Opportunities

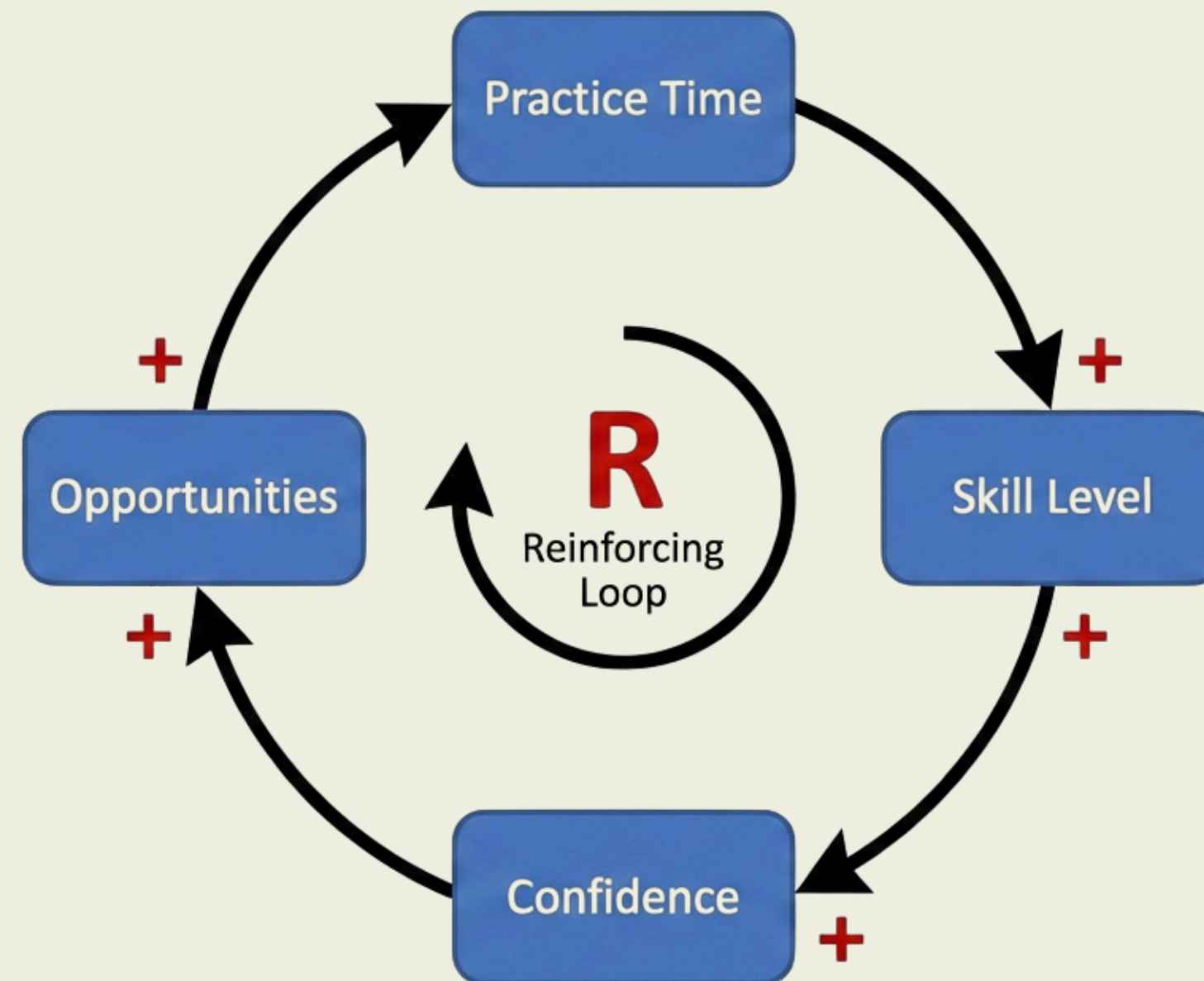
Diagram Logic

Practice Time (+) → Skill Level (+) → Confidence (+) →
Opportunities (+) → Practice Time

Example of a Causal Diagram

Learning & Skill Improvement (Reinforcing Loop)

Diagram



Example of a Causal Diagram

Learning & Skill Improvement (Reinforcing Loop)

Insight

- More practice → Better skill
- Better skill → More confidence
- More confidence → More opportunities
- More opportunities → Even more practice

This becomes a self-amplifying growth loop.